MA1522 Linear Algebra for Computing

Lesson 13: Orthogonality, and Orthogonal and Orthonormal Bases

Announcement: Online Quiz 3

- **Quiz 3 covers Chapter 3**, Sections 3.1–3.8.

- It opens this Saturday, **Saturday 3 October, 0:00**, and is due **Sunday 4 October, 23:59**.

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson13` (e.g. `A1234567Z_lesson13.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 13**.
- Deadline: today, **12:30 PM**.
- If you do not know how to upload your submission, come to ask me after class.

Pollev



Section 5.1 Orthogonality

Let $\mathbf{u} = (u_i)$ and $\mathbf{v} = (v_i)$ be vectors in $\mathbb{R}^n$.

Inner product, norm and angle

$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n = \mathbf{u}^T \mathbf{v}, \qquad \|\mathbf{u}\| = \sqrt{\mathbf{u} \cdot \mathbf{u}} = \sqrt{u_1^2 + \cdots + u_n^2}.$

In particular $\mathbf{u} \cdot \mathbf{u} \geq 0$, with equality if and only if $\mathbf{u} = \mathbf{0}$.

For $\mathbf{u}, \mathbf{v} \neq \mathbf{0}$, the **angle** between them is

$$\theta = \cos^{-1}\left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \, \|\mathbf{v}\|}\right), \qquad 0 \leq \theta \leq \pi.$$

Definition

1. $\mathbf{u}$ and $\mathbf{v}$ are **orthogonal** if $\mathbf{u} \cdot \mathbf{v} = 0$. For $\mathbf{u}, \mathbf{v} \neq \mathbf{0}$, this means the angle between them is $\theta = 90°$.
2. A set $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is **orthogonal** if $\mathbf{v}_i \cdot \mathbf{v}_j = 0$ for every $i \neq j$.
3. S is **orthonormal** if it is orthogonal and every $\mathbf{v}_i$ is a unit vector, $\|\mathbf{v}_i\| = 1$.

Example 1

True or false? Let $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_k\} \subseteq \mathbb{R}^n$.

1. If $\mathbf{w} \cdot \mathbf{v}_i = 0$ for every i, then $\mathbf{w}$ is orthogonal to every vector in $\text{span}(S)$.
2. If S is orthonormal, then S is linearly independent.

1. True. For any $c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k \in \text{span}(S)$,

$$\mathbf{w} \cdot (c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k) = c_1\,\mathbf{w} \cdot \mathbf{v}_1 + \cdots + c_k\,\mathbf{w} \cdot \mathbf{v}_k = 0.$$

2. True. Suppose $c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k = \mathbf{0}$. Take the inner product of both sides with $\mathbf{v}_i$:

$$0 = \mathbf{v}_i \cdot \mathbf{0} = c_1\,\mathbf{v}_i \cdot \mathbf{v}_1 + \cdots + c_k\,\mathbf{v}_i \cdot \mathbf{v}_k = c_i\,\mathbf{v}_i \cdot \mathbf{v}_i = c_i.$$

Section 5.1 Orthogonal Complement

Let V be a subspace of $\mathbb{R}^n$.

Definition

A vector $\mathbf{w}$ is **orthogonal to** V, written $\mathbf{w} \perp V$, if $\mathbf{w} \cdot \mathbf{v} = 0$ for every $\mathbf{v} \in V$. The **orthogonal complement** of V is

$$V^\perp = \{\, \mathbf{w} \in \mathbb{R}^n : \mathbf{w} \cdot \mathbf{v} = 0 \text{ for all } \mathbf{v} \in V \,\}.$$

Theorem

Let $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ be a spanning set for V.

1. $\mathbf{w} \perp V$ if and only if $\mathbf{w} \cdot \mathbf{u}_i = 0$ for all $i = 1, \ldots, k$.
2. Let $A = \begin{pmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_k \end{pmatrix}$. Then $V^\perp = \text{Null}(A^T)$.

Question 1

Let $V = \text{span}\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\} \subseteq \mathbb{R}^5$, where

$$\mathbf{a}_1 = \begin{pmatrix} 1 \\ 2 \\ 0 \\ 1 \\ -1 \end{pmatrix}, \qquad \mathbf{a}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \\ -1 \\ 2 \end{pmatrix}, \qquad \mathbf{a}_3 = \begin{pmatrix} 1 \\ 1 \\ -1 \\ 2 \\ -3 \end{pmatrix}.$$

Find a basis for $V^\perp$, and find $\dim(V)$ and $\dim(V^\perp)$.

Question 1 — solution

$V^\perp = \text{Null}(A^T)$, where the rows of A^T are $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$. Reduce:

$$\begin{pmatrix} 1 & 2 & 0 & 1 & -1 \\ 0 & 1 & 1 & -1 & 2 \\ 1 & 1 & -1 & 2 & -3 \end{pmatrix} \xrightarrow[R_1 - 2R_2]{R_3 - R_1,\ R_3 + R_2} \begin{pmatrix} 1 & 0 & -2 & 3 & -5 \\ 0 & 1 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

With $x_3 = s$, $x_4 = t$, $x_5 = r$: $x_1 = 2s - 3t + 5r$, $x_2 = -s + t - 2r$, so a basis for $V^\perp$ is

$$\{(2,-1,1,0,0)^T,\ (-3,1,0,1,0)^T,\ (5,-2,0,0,1)^T\}. \qquad (2,-1,1,0,0)^T \cdot \mathbf{a}_1 = 0\ \checkmark$$

The reduction has a zero row: $\mathbf{a}_3 = \mathbf{a}_1 - \mathbf{a}_2$, and $\{\mathbf{a}_1, \mathbf{a}_2\}$ is linearly independent. So $\dim(V) = \text{rank}(A^T) = 2$ and $\dim(V^\perp) = 3$.

□

Section 5.2 Orthogonal and Orthonormal Bases

Definition

Let V be a subspace of $\mathbb{R}^n$. A set S is an **orthogonal basis** (resp. **orthonormal basis**) for V if S is a basis for V and S is an orthogonal (resp. orthonormal) set.

Theorem (Coordinates Relative to an Orthogonal Basis)

Let $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ be an orthogonal basis for V. Then for every $\mathbf{v} \in V$,

$$\mathbf{v} = \frac{\mathbf{v} \cdot \mathbf{u}_1}{\|\mathbf{u}_1\|^2}\mathbf{u}_1 + \cdots + \frac{\mathbf{v} \cdot \mathbf{u}_k}{\|\mathbf{u}_k\|^2}\mathbf{u}_k, \qquad \text{so} \qquad [\mathbf{v}]_S = \begin{pmatrix} \mathbf{v} \cdot \mathbf{u}_1/\|\mathbf{u}_1\|^2 \\ \vdots \\ \mathbf{v} \cdot \mathbf{u}_k/\|\mathbf{u}_k\|^2 \end{pmatrix}.$$

If S is orthonormal, $[\mathbf{v}]_S = (\mathbf{v} \cdot \mathbf{u}_1, \ldots, \mathbf{v} \cdot \mathbf{u}_k)^T$.

Question 2

Let $V = \text{span}(S)$, where $S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ and

$$\mathbf{u}_1 = \begin{pmatrix} 1 \\ -1 \\ 1 \\ 1 \end{pmatrix}, \qquad \mathbf{u}_2 = \begin{pmatrix} 1 \\ 1 \\ -1 \\ 1 \end{pmatrix}, \qquad \mathbf{u}_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ -1 \end{pmatrix}, \qquad \mathbf{v} = \begin{pmatrix} 2 \\ -2 \\ 0 \\ 4 \end{pmatrix}.$$

Show that S is an orthogonal basis for V. Then, given that $\mathbf{v} \in V$, find $[\mathbf{v}]_S$ without solving a linear system.

Question 2 — solution

Put $A = (\,\mathbf{u}_1\ \mathbf{u}_2\ \mathbf{u}_3\ \mathbf{v}\,)$. The (i,j)-entry of A^TA is the inner product of columns i and j of A:

$$A^TA = \begin{pmatrix} 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & -1 \\ 2 & -2 & 0 & 4 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 & 2 \\ -1 & 1 & 1 & -2 \\ 1 & -1 & 1 & 0 \\ 1 & 1 & -1 & 4 \end{pmatrix} = \begin{pmatrix} 4 & 0 & 0 & 8 \\ 0 & 4 & 0 & 4 \\ 0 & 0 & 4 & -4 \\ 8 & 4 & -4 & 24 \end{pmatrix}.$$

The top-left 3×3 block is diagonal with nonzero diagonal: S is an orthogonal set of nonzero vectors, hence linearly independent (as in Example 1, part 2), and it spans V. So S is an orthogonal basis for V.

The last column gives $\mathbf{v} \cdot \mathbf{u}_i = 8, 4, -4$, and $\|\mathbf{u}_i\|^2 = 4$. By the formula,

$$[\mathbf{v}]_S = \begin{pmatrix} 8/4 \\ 4/4 \\ -4/4 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}. \qquad 2\mathbf{u}_1 + \mathbf{u}_2 - \mathbf{u}_3 = (2, -2, 0, 4)^T = \mathbf{v}. \checkmark$$

□

Challenge 1

Let $V = \text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ be as in Question 2. For $\mathbf{w} \in \mathbb{R}^4$, let $c_i = \mathbf{u}_i \cdot \mathbf{w}$ for $i = 1, 2, 3$, and set

$$\mathbf{w}' = \mathbf{w} - \left(\frac{c_1}{\|\mathbf{u}_1\|^2}\mathbf{u}_1 + \frac{c_2}{\|\mathbf{u}_2\|^2}\mathbf{u}_2 + \frac{c_3}{\|\mathbf{u}_3\|^2}\mathbf{u}_3 \right).$$

1. Show that $\mathbf{w}' \in V^{\perp}$.
2. Show that $\mathbf{w} \in V$ if and only if $\mathbf{w}' = \mathbf{0}$.

Section 5.2 Orthogonal Matrices

Let $Q = (\mathbf{u}_1\ \mathbf{u}_2\ \cdots\ \mathbf{u}_k)$ be an $n \times k$ matrix. The (i,j)-entry of Q^TQ is $\mathbf{u}_i \cdot \mathbf{u}_j$, so $\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ is orthonormal if and only if $Q^TQ = I_k$.

Definition

A square matrix A of order n is **orthogonal** if $A^T = A^{-1}$, equivalently, $A^TA = I_n = AA^T$.

Theorem

Let A be a square matrix of order n. The following are equivalent.

(i) A is an orthogonal matrix.

(ii) The columns of A form an orthonormal basis for $\mathbb{R}^n$.

(iii) The rows of A form an orthonormal basis for $\mathbb{R}^n$.

Question 3

You may assume that $x^2 + y^2 = 1$ if and only if $x = \cos\theta$ and $y = \sin\theta$ for some θ.

1. Let $\mathbf{u} = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$. Suppose that the 2×2 matrix $(\,\mathbf{u}\ \mathbf{v}\,)$ is orthogonal. Find all possible $\mathbf{v}$.

2. Let $\mathbf{u} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$. Suppose that the 3×3 matrix $(\,\mathbf{u}\ \mathbf{v}\ \mathbf{w}\,)$ is orthogonal. Find all possible $\mathbf{v}$ and $\mathbf{w}$.

Question 3 — solution

By the Theorem, a square matrix is orthogonal if and only if its columns form an orthonormal set.

1. We need $\mathbf{v} \cdot \mathbf{u} = v_1 \cos\theta + v_2 \sin\theta = 0$ and $\|\mathbf{v}\| = 1$. Since $\mathbf{u} \neq \mathbf{0}$, the solutions of the first equation are $\mathbf{v} = s \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix}$, and then $\|\mathbf{v}\| = |s| = 1$. So

$$\mathbf{v} = \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix} \quad \text{or} \quad \mathbf{v} = \begin{pmatrix} \sin\theta \\ -\cos\theta \end{pmatrix}.$$

2. $\mathbf{v} \cdot \mathbf{u} = v_3 = 0$ and $v_1^2 + v_2^2 = 1$, so $\mathbf{v} = \begin{pmatrix} \cos\varphi \\ \sin\varphi \\ 0 \end{pmatrix}$ for some φ. Likewise $w_3 = 0$, and $\begin{pmatrix} w_1 \\ w_2 \end{pmatrix}$ is a unit vector orthogonal to $\begin{pmatrix} \cos\varphi \\ \sin\varphi \end{pmatrix}$, so by part 1,

$$\mathbf{v} = \begin{pmatrix} \cos\varphi \\ \sin\varphi \\ 0 \end{pmatrix}, \qquad \mathbf{w} = \pm \begin{pmatrix} -\sin\varphi \\ \cos\varphi \\ 0 \end{pmatrix}.$$

□

Challenge 2

Let A and B be orthogonal matrices of order n.

1. Show that $\det(A) = 1$ or $\det(A) = -1$.
2. Show that AB is an orthogonal matrix.
3. Show that A^{-1} is an orthogonal matrix.

Next lesson

Next lesson (**Fri 2 Oct**, Sections 5.2 & 5.3), I will focus on

1. the definition of orthogonal projection, and how to compute it;

2. how to perform Gram-Schmidt Orthogonalization to transform a basis into an orthonormal basis, on an explicit example.

Please watch the video clips beforehand.

Challenge 1 — solution

1. Since $\mathbf{u}_j \cdot \mathbf{u}_i = 0$ for $i \neq j$, only the $i = j$ term survives: for $j = 1, 2, 3$,

$$\mathbf{u}_j \cdot \mathbf{w}' = \mathbf{u}_j \cdot \mathbf{w} - \sum_{i=1}^{3} \frac{c_i}{\|\mathbf{u}_i\|^2} \mathbf{u}_j \cdot \mathbf{u}_i = c_j - \frac{c_j}{\|\mathbf{u}_j\|^2} \|\mathbf{u}_j\|^2 = 0.$$

Since $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$ span V, $\mathbf{w}' \in V^\perp$ (Example 1, part 1).

2. ($\Leftarrow$) If $\mathbf{w}' = \mathbf{0}$, then $\mathbf{w} = \sum_i \frac{c_i}{\|\mathbf{u}_i\|^2} \mathbf{u}_i \in V$.

($\Rightarrow$) If $\mathbf{w} \in V$, then since S is an orthogonal basis for V, the coordinate formula gives

$$\mathbf{w} = \frac{\mathbf{w} \cdot \mathbf{u}_1}{\|\mathbf{u}_1\|^2} \mathbf{u}_1 + \frac{\mathbf{w} \cdot \mathbf{u}_2}{\|\mathbf{u}_2\|^2} \mathbf{u}_2 + \frac{\mathbf{w} \cdot \mathbf{u}_3}{\|\mathbf{u}_3\|^2} \mathbf{u}_3 = \frac{c_1}{\|\mathbf{u}_1\|^2} \mathbf{u}_1 + \frac{c_2}{\|\mathbf{u}_2\|^2} \mathbf{u}_2 + \frac{c_3}{\|\mathbf{u}_3\|^2} \mathbf{u}_3,$$

that is, $\mathbf{w}' = \mathbf{0}$. □

Challenge 2 — solution

1. Since $A^TA = I$ and $\det(A^T) = \det(A)$,

$$1 = \det(I) = \det(A^TA) = \det(A^T)\det(A) = \det(A)^2.$$

Hence $\det(A) = \pm 1$. Both values occur: in Question 3 part 1, $\det\begin{pmatrix}\cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{pmatrix} = 1$ and $\det\begin{pmatrix}\cos\theta & \sin\theta \\ \sin\theta & -\cos\theta\end{pmatrix} = -1$.

2. AB is square of order n, and

$$(AB)^T(AB) = B^TA^TAB = B^TIB = B^TB = I.$$

So AB is orthogonal.

3. $A^{-1} = A^T$, and $(A^T)^TA^T = AA^T = I$. So A^{-1} is orthogonal. □